

YOGESH KUCHIMANCHI

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PROFILE

M.S. Data Science candidate and researcher building reproducible ML/NLP systems with Python, SQL, PyTorch, and Streamlit. Analyzed 351K+ records, fine-tuned Qwen3.5-9B with LoRA, and authored a paper accepted at ASONAM 2026.

EDUCATION

Rochester Institute of Technology <i>M.S. in Data Science Machine Learning, Deep Learning, Cloud Computing, Big Data Analytics</i>	Aug 2024 - Expected Dec 2026 Rochester, NY
Vellore Institute of Technology <i>Postgraduate Program in Data Science</i>	Aug 2023 - Jul 2024 India
New Shores International College <i>B.C.A. in Computer Applications</i>	Nov 2020 - Dec 2023 India

EXPERIENCE

Graduate Researcher, Data Science Rochester Institute of Technology Women's Safety Narratives	2025 - 2026
- Built Python/Pandas pipelines to analyze 351,501 Reddit and YouTube comments across 16 women-safety cases in India (2012-2024), producing evidence for policy recommendations. - Applied Mann-Whitney U, chi-square, and G-tests; fine-tuned Qwen3.5-9B with LoRA and evaluated 3,000 held-out examples (0.745 macro-F1; 0.663 MCC). - Authored a paper accepted at ASONAM 2026 and documented model limitations through high-confidence error review.	
GCCIS Technical Assistant Rochester Institute of Technology	Aug 2025 - Present
Support 200+ Windows/Linux lab systems, troubleshoot access and software issues, and maintain Excel/VBA reports for faculty and students.	
Research Assistant, Data Science Rochester Institute of Technology	Dec 2024 - Apr 2025
Refactored Pandas/NumPy scripts and validated PostgreSQL/Excel data, supporting accurate weekly research reporting throughout a five-month appointment.	

PROJECTS

Clinical Readmission Risk Modeling, Calibration & Drift <i>Python, CatBoost, PyTorch, Streamlit</i> GitHub	
• Built a leakage-safe binary 30-day readmission pipeline on 83,621 UCI hospital encounters with patient-disjoint, encounter-order train/validation/test cohorts and zero patient overlap. • Benchmarked linear, tree, and neural models; retained calibrated CatBoost at 0.661 ROC-AUC, 0.175 PR-AUC, and 0.077 Brier score on 10,822 held-out rows; added PSI/KS/JS drift audits. • Fit preprocessing on training data, selected thresholds and calibration on validation only, preserved test for final evaluation, built a Streamlit audit dashboard, and covered pipeline logic with 12 automated tests.	
Transformer News Classification <i>DistilBERT, FastAPI, React</i> Live Demo GitHub	
• Fine-tuned DistilBERT on 96,000 AG News records for four-class classification; evaluated on 12,000 held-out articles (0.870 accuracy, 0.869 macro-F1, 0.827 MCC). • Built FastAPI inference with schema validation and request throttling, a React confidence dashboard, Docker packaging, CI/CD workflows, and 22 automated tests. • Deployed the public dashboard to Vercel and clearly separated hosted demo-mode inference from local real-model inference.	
Model Behavior Under Distribution Shift <i>Python, NumPy, Pandas, Streamlit</i> GitHub	
• Built a simulator that trains a NumPy logistic model and monitors ROC-AUC/F1, class-prior shift, PSI, approximate KS, and configurable alert thresholds. • Verified a mixed-shift run on 4,000 reference and 4,000 current rows: ROC-AUC fell from 0.758 to 0.372, max PSI reached 0.608, and three alerts fired. • Covered deterministic data generation and monitoring calculations with three automated tests.	

TECHNICAL SKILLS

Languages: Python, SQL, R, Java
Machine Learning & NLP: PyTorch, scikit-learn, CatBoost, XGBoost, Hugging Face Transformers, LoRA, DistilBERT, FT-Transformer
Statistics & Evaluation: A/B testing, Mann-Whitney U, chi-square, G-test, ROC-AUC, PR-AUC, probability calibration, PSI/KS/JS drift analysis
Data & Databases: pandas, NumPy, SciPy, Spark, PostgreSQL, SQLite, SQL CTEs, joins, aggregations, window functions
ML Engineering & Apps: FastAPI, Streamlit, React, Docker, GitHub Actions, CI/CD, model serving, automated testing, Git, uv, Excel/VBA